

Extension: Geometric mean



1

a 15

b 30

2

a 15.72

b 7.76

3

a 20 m

b 17.861 mm

4

a 21 mm

b 18.2326 m

5 $\sqrt{24}$

6 21.3 units

7 2.0

8 100 cm

9 $27\sqrt{3}$ m

10

a 26.0

b $\$I$ plus the constant interest rate that has the same return after three years on an initial investment of $\$I$.

11

a 36.33

b 10.96

c

Applicant	Arithmetic mean	Geometric mean
Beth	36.33	10.96
Isabelle	33.67	19.63
Ursula	35.00	18.52

d Isabelle



12

a

	Arithmetic Mean	Geometric Mean
Town 1	11	4
Town 2	9	9

b $x = 23$

c $y = 2396$

13

a

i $\frac{35}{2}$ cm

ii $5\sqrt{6}$ cm

b

i Yes

ii No

iii Yes

iv No

c

i $\frac{l+w}{2}$ cm

ii $\sqrt{lw}$ cm

d

i Yes

ii No

iii No

iv No